

Отчёт к лабораторной работе 6

Тема – «Создание базы данных в MongoDB»

База данных:

Создадим через json:

```
[
  {
    "_id": 1,
    "surname": "Оборин",
    "name": "Сергей",
    "patronymic": "Иванович",
    "email_address": "seryi666@mail.ru",
    "phone_number": "+79528123535"
  },
  {
    "_id": 2,
    "surname": "Depp",
    "name": "Johnny",
    "patronymic": null,
    "email_address": "depp@google.com",
    "phone_number": "+7777777777"
  },
  {
    "_id": 3,
    "surname": "Маятин",
    "name": "Александр",
    "patronymic": "Владимирович",
    "email_address": "mayatin@itmo.ru",
    "phone_number": "+78124801290"
  },
  {
    "_id": 4,
    "surname": "Сидякин",
    "name": "Ярослав",
    "patronymic": "Андреевич",
    "email_address": "хоч1404@yahoo.com",
    "phone_number": "+79177657782"
  },
  {
    "_id": 5,
    "surname": "Сидоров",
    "name": "Иван",
    "patronymic": "Петрович",
    "email_address": "sip@mail.ru",
    "phone_number": "+78005553535"
  },
  {
    "_id": 6,
    "number": "N4-254",
    "departure_date": {
      "$date": "2024-03-19T14:55:00Z"
    },
    "arrival_date": {
      "$date": "2024-03-19T16:00:00Z"
    },
    "departure_airport_id": 1,
    "arrival_airport_id": 2,
    "airplane_id": 1
  },
  {
    "_id": 7,
    "number": "M7-250",
    "departure_date": {
      "$date": "2024-03-23T15:55:00Z"
    },
    "arrival_date": {
      "$date": "2024-03-23T17:35:00Z"
    },
    "departure_airport_id": 3,
    "arrival_airport_id": 5,
    "airplane_id": 4
  },
  {
    "_id": 8,
    "number": "M5-008",
    "departure_date": {
      "$date": "2024-03-20T10:00:00Z"
    },
    "arrival_date": {
      "$date": "2024-03-20T11:20:00Z"
    },
    "departure_airport_id": 4,
    "arrival_airport_id": 2,
    "airplane_id": 5
  },
  {
    "_id": 9,
    "number": "A7-404",
    "departure_date": {
      "$date": "2024-03-19T12:25:00Z"
    },
    "arrival_date": {
      "$date": "2024-03-19T14:10:00Z"
    },
    "departure_airport_id": 5,
    "arrival_airport_id": 1,
    "airplane_id": 1
  },
  {
    "_id": 10,
    "flight_id": 3,
    "booking_date": {
      "$date": "2024-03-17T16:04:14Z"
    },
    "users": [
      {
        "user_id": 1,
        "place_number": "4A"
      },
      {
        "user_id": 2,
        "place_number": "4B"
      }
    ]
  },
  {
    "_id": 11,
    "flight_id": 3,
    "booking_date": {
      "$date": "2024-03-14T17:35:03Z"
    },
    "users": [
      {
        "user_id": 5,
        "place_number": "10A"
      }
    ]
  },
  {
    "_id": 12,
    "flight_id": 2,
    "booking_date": {
      "$date": "2024-03-15T11:15:35Z"
    },
    "users": [
      {
        "user_id": 5,
        "place_number": "5F"
      }
    ]
  },
  {
    "_id": 13,
    "flight_id": 4,
    "booking_date": {
      "$date": "2024-03-12T14:27:56Z"
    },
    "users": [
      {
        "user_id": 4,
        "place_number": "7E"
      }
    ]
  },
  {
    "_id": 14,
    "flight_id": 1,
    "booking_date": {
      "$date": "2024-03-16T15:09:14Z"
    },
    "users": [
      {
        "user_id": 4,
        "place_number": "7E"
      }
    ]
  }
]
```

airplane Storage size: 20.48 kB Documents: 10 Avg. document size: 53.00 B Indexes: 1 Total index size: 20.48 kB	airport Storage size: 20.48 kB Documents: 10 Avg. document size: 112.00 B Indexes: 1 Total index size: 20.48 kB	booking Storage size: 20.48 kB Documents: 10 Avg. document size: 113.00 B Indexes: 1 Total index size: 20.48 kB	flight Storage size: 20.48 kB Documents: 10 Avg. document size: 147.00 B Indexes: 1 Total index size: 20.48 kB
payment Storage size: 20.48 kB Documents: 10 Avg. document size: 42.00 B Indexes: 1 Total index size: 20.48 kB	user Storage size: 20.48 kB Documents: 10 Avg. document size: 166.00 B Indexes: 1 Total index size: 20.48 kB		

Создание индекса:

arrival_date_1 departure_date_1

REGULAR ⓘ

20.5 KB

0 (since Sat May 31 2025)

COMPOUND ⓘ

READY

arrival_date ↑ departure_date ↑

Создание представления:

```

db.createView(
  "booking_flight_details",
  "booking",
  [
    {
      $lookup: {
        from: "flight",
        localField: "flight_id",
        foreignField: "_id",
        as: "flight"
      }
    },
    {
      $unwind: "$flight"
    }
  ]
)

```

airplane Storage size: 20.48 kB Documents: 10 Avg. document size: 53.00 B Indexes: 1 Total index size: 36.86 kB	airport Storage size: 20.48 kB Documents: 10 Avg. document size: 112.00 B Indexes: 1 Total index size: 36.86 kB	booking Storage size: 20.48 kB Documents: 10 Avg. document size: 113.00 B Indexes: 1 Total index size: 36.86 kB	booking_flight_details VIEW READ-ONLY View on: booking
flight Storage size: 20.48 kB Documents: 10 Avg. document size: 147.00 B Indexes: 2 Total index size: 57.34 kB	payment Storage size: 20.48 kB Documents: 10 Avg. document size: 42.00 B Indexes: 1 Total index size: 36.86 kB	user Storage size: 20.48 kB Documents: 10 Avg. document size: 166.00 B Indexes: 1 Total index size: 36.86 kB	

```
> show collections
< airplane
  airport
  booking
  booking_flight_details [view]
  flight
  payment
  user
  system.views
> db.booking_flight_details.find().pretty()
< {
  _id: 1,
  flight_id: 3,
  booking_date: 2024-03-17T16:04:14.000Z,
  users: [
    {
      user_id: 1,
      place_number: '4A'
    },
    {
      user_id: 2,
      place_number: '4B'
    }
  ],
  flight: {
    _id: 3,
    number: 'M5-008',
    departure_date: 2024-03-20T10:00:00.000Z,
    arrival_date: 2024-03-20T11:20:00.000Z,
    departure_airport_id: 4,
    arrival_airport_id: 2,
    airplane_id: 5
  }
}
```